

Typhoidal *Salmonella*

**Background document for the
WHO Guidelines for drinking-water quality and the
WHO Guidelines on sanitation and health**

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Highlights

- There is strong evidence linking transmission of typhoidal *Salmonella* with unsafe drinking-water, inadequate sanitation, and poor hygiene. *Salmonella* Typhi and *S. Paratyphi* are major waterborne pathogens. Multiple drinking-water outbreaks have been documented and there is also evidence from endemic typhoid and paratyphoid fever.
- *Salmonella* Typhi and *S. Paratyphi* are specific to human hosts.
- Transmission is primarily by the faecal–oral route. Water and food are the primary sources of infection. Direct person-to-person transmission is less common.
- Initial symptoms of *S. Typhi* infection (typhoid fever) are headache, fever, diffuse abdominal pain, and malaise, but cases become progressively unwell without treatment, resulting in severe disease. Complications include intestinal haemorrhage and perforation and may result in death. Between 1% and 5% of cases become chronic carriers. Infection with *S. Paratyphi* causes similar, although generally milder, symptoms.
- Multiple antibiotic resistance to typhoidal *Salmonella* emerged in the 1980s, and increasing levels of antimicrobial resistance to fluoroquinolones (e.g. ciprofloxacin) are being reported.
- In endemic areas, typhoid fever is most common in children. In low prevalence areas, incidence of typhoid fever has a more even age distribution, and a high proportion of cases are a result of travel to endemic disease areas.
- Relative infectivity of *S. Typhi* is low with a calculated ID₅₀ (50% infective dose) of 1.1×10^6 organisms. A not fully effective form of immunity may follow recovery from symptomatic and asymptomatic infection.
- Culture-based methods, serological classification and molecular techniques are available for detection and identification in environmental samples.
- There are limited data on occurrence and prevalence of *S. Typhi* and *S. Paratyphi* in water environments. Typhoidal *Salmonella* exhibit short persistence in fresh water.
- Systematic approaches that identify and control sources of faecal contamination along the whole sanitation service and drinking-water supply chains are needed to manage microbial risks, including from typhoidal *Salmonella*.
- Reduction of *Salmonella* by wastewater and drinking-water treatment varies depending on the process but is similar to reduction of faecal indicator bacteria such as *E. coli*. Disinfection processes including chlorination, are effective for inactivating typhoidal *Salmonella* in drinking-water.
- *Escherichia coli* (or alternatively, thermotolerant coliforms) is a suitable indicator for the possible presence or absence of typhoidal *Salmonella* in drinking-water.

1 General description

The genus *Salmonella* belongs to the family Enterobacteriaceae. *Salmonella* are predominantly motile, Gram-negative rods (0.7 µm to 1.5 µm wide; 2 µm to 5 µm long), non-spore forming, facultative aerobes, with peritrichous flagellae (McQuiston et al., 2008). *Salmonella* can survive, and even grow, in a diverse range of stressful environments including: extracellular pH from 4 to 9.5, salt concentrations up to 4% NaCl and temperatures as high as 54 °C or low as 2 °C (D'Aoust et al., 2001; Levantesi et al., 2012). The optimal growth temperature range for *Salmonella* is 35 °C to 43 °C. There has been much debate about nomenclature and taxonomy of *Salmonella*, but it is now considered that there are two species, *Salmonella enterica* and *Salmonella bongori* (Tindall et al., 2005; McQuiston et al., 2008). *Salmonella enterica* (subspecies enterica, salamae, arizonae, diarizonae, indica and houtenae) are associated with greater than 99% of *Salmonella*-caused diseases in humans and warm-blooded animals (McQuiston et al., 2008; Chen et al., 2013).

In addition to classification of subspecies, *Salmonella* are further subdivided into serovars according to their flagellar (h) and somatic (o) antigens and have different host specificity, diverse geographic distribution, and cause different syndromes. *Salmonella enterica* subspecies enterica serovar Typhi (*S. Typhi*) and serovar Paratyphi (*S. Paratyphi*) are known as the typhoidal *Salmonella* for their ability to cause typhoid and paratyphoid fevers in humans respectively, whereas all other *Salmonella* are non-typhoidal (Antunes et al., 2015; Bell et al., 2016). *Salmonella* Paratyphi can be further subdivided into serotypes A, B and C. *Salmonella* Paratyphi A is the most common cause of paratyphoid fever while *S. Paratyphi* C is extremely rare. Non-typhoidal *Salmonella* are the subject of a separate background document.

2 Human health effects

In endemic areas many of those infected, particularly children under 5 years of age may have a mild, non-specific gastrointestinal disease which is not recognized clinically as typhoid fever. Asymptomatic infection can also occur (Parry et al., 2002; Heymann, 2015). About 10% to 15% of patients develop severe disease (Bhan et al., 2005; Crump et al., 2015); if untreated, illness typically lasts 4 weeks. Initially symptoms are non-specific and may include headache, malaise, cough, diffuse abdominal pain, constipation and remittent fever (Crump et al., 2015; Hasan et al., 2019). As the disease progresses intermittent confusion and apathy with sustained fever may be experienced. During the third week the patient becomes progressively more unwell with a continuous high temperature and mental confusion. Abdominal distention can be marked, and some patients pass a foul yellow-green “pea soup” stool. During the third week the risk of serious complications and death is highest. The most serious and life-threatening complications are intestinal haemorrhage and perforation. If the patient survives to the fourth week the fever slowly resolves and the mental state and abdominal distention improve. However, continuing weakness and weight loss may persist for months. The classical presentation of typhoid fever is rare in high-income countries since diagnosis and antimicrobial therapy halts progression. Various atypical presentations of typhoid fever are seen including pneumonia, joint infections or urinary tract infections (Heymann, 2015).

Multi-drug resistance to ampicillin, chloramphenicol, and trimethoprim-sulfamethoxazole emerged in the 1980s (Date et al., 2016; Radhakrishnan et al., 2018) and increasing levels of

antimicrobial resistance to fluoroquinolones (e.g. ciprofloxacin) are being reported (Wain et al., 2014; Shresthra et al., 2015).

3 Epidemiology

Understanding the global distribution of typhoid and paratyphoid fever is undermined by weak surveillance systems in regions where the burden is likely to be greatest (Buckle et al., 2012; Wain et al., 2014; Radhakrishnan et al., 2018). It is estimated that globally, there are 11 million to 21 million typhoid and paratyphoid cases and 128 000 to 161 000 deaths each year (WHO, 2018). The highest rates of infection are in south-east Asia and sub-Saharan Africa (Steinberg et al., 2004; Wain et al., 2014; Radhakrishnan et al., 2018; Hasan et al., 2019). Although typhoid fever is more prevalent than paratyphoid fever, reports suggest that *S. Paratyphi A* infection is increasing in some areas of Asia (Wain et al., 2014).

The age distribution of cases varies between endemic and low prevalence settings. In endemic areas the disease is most common in children (WHO, 2018). Host factors associated with increased risk of infection or severe infection include genetic traits, recent use of antimicrobial medications and presence of other infections (Mitscherlich & Marth, 1984; Ruby et al., 2012), although immunocompromised patients do not appear to be more susceptible to infection (Gordon, 2008). In low prevalence areas, incidence of typhoid fever has a more even age distribution (Crump et al., 2004), and a high proportion of cases are a result of travel to endemic disease areas (Date et al., 2016).

4 Transmission

Salmonella Typhi and *S. Paratyphi* are spread primarily by the faecal–oral (or urine–oral) route. Unlike non-typhoidal *Salmonella*, which can infect a wide range of animals, *S. Typhi* and *S. Paratyphi* are largely restricted to humans (Hasan et al., 2019). Infection by typhoidal *Salmonella* is mostly associated with the consumption of faecally contaminated water or food (Parry et al., 2002; WHO, 2018). There is a strong association of typhoid and paratyphoid fever with inadequate sanitation, unsafe drinking-water and poor hygiene (Crump & Mintz, 2010; WHO, 2018). In endemic areas, there is evidence that children contract *S. Typhi* from heavily contaminated environments through physical contact with water sources (Akullian et al., 2015). Typhoid fever can occasionally be transmitted by direct person-to-person contact, but this is a less important route than contaminated food or water (Bhan et al., 2005; WHO, 2018).

The relative infectivity of *S. Typhi* is low¹ with an ID₅₀ (50% infective dose i.e. the number of organisms required to infect 50% of exposed subjects) of 1.1×10^6 organisms (Hornick et al., 1970). The incubation period for typhoid fever can be 3 days to more than 60 days but is usually 8 to 14 days. For paratyphoid fever, the incubation period is slightly shorter at 1 to 10 days (Heymann, 2015). Patients excrete the pathogen throughout the period of symptomatic infections and for variable periods afterwards, commonly 1 to 2 weeks for paratyphoid. About

¹ From experiments with human volunteers, from epidemiological evidence and from experimental animal studies. Based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: High means infective doses can be $1-10^2$ organisms or particles, Moderate 10^2-10^4 and Low $> 10^4$.

10% of typhoid fever patients with no treatment continue to shed the pathogens for 3 months after onset of symptoms. A not fully effective form of immunity may follow recovery from symptomatic and asymptomatic infection (Heymann, 2015). Between 1% to 5% of patients become chronic carriers in most cases without ongoing symptoms and excrete *S. Typhi* or *S. Paratyphi* in urine or stools for more than a year (Bhan et al., 2005). Asymptomatically infected individuals can also shed pathogens, and almost 25% of chronic carriers will not exhibit typhoid fever (Levine et al., 1982). Chronic carriage is higher in women, those over 50 years of age and patients with certain other underlying diseases (Levine et al., 1982).

5 Detection methods

Standard culture-based methods for identification of *S. Typhi* and *S. Paratyphi* in environmental samples are available (e.g. ISO, 2010), and serological typing methods are available to confirm their presence (Bell et al., 2016). Methods for detection and confirmation include a number of steps and take several days to complete. For quantification most probable number (MPN) tests can be performed.

Immunoassays and molecular methods are also available for the detection of typhoidal *Salmonella* in water (Karkey et al., 2016). Distinguishing between viable and non-viable cells is a shortcoming of molecular methods.

6 Source and occurrence

Occurrence data for *S. Typhi* and *S. Paratyphi* are relatively scarce. There are much more data on occurrence of non-typhoidal *Salmonella*.

6.1 Faecal pollution sources

Salmonella Typhi and *S. Paratyphi* have been detected in sewage (Thong et al., 1996; Yan et al., 2018), but limited data are available on concentrations or survival. It has been estimated that faeces can contain 10^6 to 10^{10} organisms/g (Gopinath et al., 2012). No data were identified on occurrence in sludge and treated wastewater.

6.2 Surface water

Salmonella Typhi and *S. Paratyphi* A have been detected in surface waters by culture and molecular methods in countries where typhoid fever is endemic (Thong et al., 1996; Levantesi et al., 2012) but limited data are available on concentrations or survival. The less common cause of typhoidal disease, *S. Paratyphi* B, was detected in surface water in France (Baudart et al., 2000). In laboratory studies starting with a concentration of 8×10^6 colony forming units (CFU)/mL, *S. Typhi* concentrations showed short persistence² with concentrations reduced by 1.3 log₁₀ per day in pond water (Cho & Kim, 1999). However, the decline in the number of viable cells as measured using a chromosomal marker was much lower at 0.38 log₁₀ per day.

² Based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: Detection period for infective stage in water at 20 °C: Short, up to 1 week; Moderate, 1 week to 1 month; Long, over 1 month.

6.3 Groundwater

In laboratory studies, starting with a concentration of 8×10^6 CFU/mL, *S. Typhi* showed short persistence with concentrations reduced by $0.75 \log_{10}$ per day in groundwater samples (Cho & Kim, 1999). However, the decline in the number of viable cells as measured using a chromosomal marker was lower at $0.28 \log_{10}$ per day. In in situ groundwater microcosms (starting concentrations of 1.5×10^7 *S. Typhi* CFU/mL), densities were reduced by $0.91 \log_{10}$ per day by culture, while the number of viable cells as measured using a chromosomal marker was reduced by $0.31 \log_{10}$ per day (Cho & Kim, 1999).

6.4 Drinking-water

Salmonella Typhi and *S. Paratyphi A* have been detected in drinking-water by culture and molecular methods in low-income countries (Karkey et al., 2016; Stella et al., 2018). Both *S. Typhi* and *S. Paratyphi* were reported in low concentrations (0–50 CFU/100 mL) in boreholes and reservoirs used for drinking in Nigeria (Stella et al., 2018). In Nepal, 77% and 70% of drinking-water samples tested ($n = 432$) were positive for *S. Typhi* and *S. Paratyphi*, respectively (Karkey et al., 2016).

6.5 Soil and crops

A 2010 outbreak of paratyphoid fever involving 601 cases in Yuanjiang County, China was linked to consumption of raw vegetables irrigated with hospital wastewater and other surface waters (Yan et al., 2015).

6.6 Surfaces

Salmonella Typhi can survive from 6 hours to 4 weeks on inanimate surfaces (Mitscherlich & Marth, 1984).

7 Significance of sanitation

Safe sanitation is an important barrier against spread of typhoidal *Salmonella* (Hasan et al., 2019). Sewage has been identified as a source of a drinking-water outbreak caused by typhoidal *Salmonella* (Qamar et al., 2018). A sanitation service chain that separates human faeces from the environment and from human contact should effectively interrupt the transmission of *S. Typhi* and *S. Paratyphi* (Crump & Mintz 2010; Akullian et al., 2015).

8 Significance in drinking-water

S. Typhi and *S. Paratyphi* are major waterborne pathogens. There is strong evidence linking unsafe drinking-water and endemic and epidemic typhoid and paratyphoid fever with multiple documented outbreaks (Luby et al., 1998; Ram et al., 2007; Levantesi et al., 2012; Prasad et al., 2018; Qamar et al., 2018). During a 2008 outbreak in Colorado USA, caused by contamination of the municipal water supply, 434 cases of *S. Typhi* and one death were reported (Ailes et al., 2013). An outbreak of *S. Typhi* in Pakistan in 2016–2017, resulting in 486 illnesses, was attributed to drinking-water mixed with sewage (Qamar et al., 2018).

9 Prevention and management

Systematic approaches that identify and control faecal contamination along the whole sanitation service and drinking-water supply chains are needed to manage microbial risks, including from typhoidal *Salmonella*.

The effectiveness of sanitation and drinking-water treatment systems depends on the selection and operation of technologies and the characteristics of the target organisms. Treatment processes based on size exclusion (e.g. filtration through media or soil in septic tank drainage areas) or settling in primary wastewater treatment processes will be limited for removing bacteria, including *Salmonella*. Oxidizing chemical disinfectants, such as chlorine, and UV light, are effective at inactivating *Salmonella* and other non-spore forming bacteria.

9.1 Sanitation systems

Sanitation safety plans (WHO, 2022) should include control measures to safely collect, treat and discharge or reuse wastewater and faecal sludges. Removal or inactivation of *Salmonella* by wastewater treatment processes is variable, although the data are limited and generally based on non-typhoidal strains. Based on available evidence, the reduction of *Salmonella* by conventional municipal wastewater treatment processes (1 to 3 log₁₀) is likely to be similar to the reduction of faecal indicator bacteria such as *E. coli* (Koivunen et al., 2003; Hasan et al., 2019). Tertiary treatment including disinfection is needed to achieve efficient reduction or a total removal of *Salmonella* (Koivunen et al., 2003; El Boulani et al., 2017).

Little data are available on the survival of *Salmonella* in latrines and septic tanks; however, the extent of removal of *Salmonella* in these systems is likely similar to that of *E. coli* (Hasan et al., 2019). Reductions in these systems generally rely on adsorption to and filtration by soil and are typically greater for bacteria than other pathogens due to their metabolic activities (Orner et al., 2018).

Anaerobic digestion of sludges produces inconsistent reductions of *Salmonella*, but removal generally improves at higher temperatures (Hasan et al., 2019). Fly larvae composting as a faecal sludge treatment method has been reported to provide substantial removal of *Salmonella* compared to controls. Lalander et al. (2013) observed a 6 log₁₀ reduction in *S. Typhimurium*/g of faeces after 8 days of larvae treatment compared to only a 1 log₁₀ reduction by storage.

9.2 Drinking-water supplies

Within a water safety plan (WHO, 2012; 2023), control measures to manage potential risks from *S. Typhi* and *S. Paratyphi* should include protection of source waters (surface or groundwater) from contamination by human faecal waste (including extra vigilance during heavy rain), application of adequate water treatment processes and protection of drinking-water quality during storage and distribution. While there are no specific data on the sensitivity of *S. Typhi* and *S. Paratyphi* to inactivation by chlorine, laboratory experiments with non-

typhoidal salmonellae strains showed they possess low resistance³. In one study Ct (product of disinfectant concentration and contact time) values of 0.4 to 0.5 mg.min/L at 25°C and pH 7 achieved a 2 log₁₀ inactivation of *S. typhimurium* (King et al., 1988). In another study of 4 NTS strains, a Ct value of less than or equal to 0.25 mg.min/L achieved a 4.5 log₁₀ inactivation of three strains and a 1.5 log₁₀ inactivation of the remaining strain (Shen et al., 2013). *Salmonella* Typhi is sensitive to UV light disinfection, i.e. 2 log₁₀ reduction at 12 mJ/cm² (Hijnen et al., 2006).

Escherichia coli (or alternatively, thermotolerant coliforms) is a suitable indicator for the possible presence or absence of *S. Typhi* and *S. Paratyphi* in drinking-water.

10 ICD-11 classification

XN0QE	<i>Salmonella</i>
XN2DW	<i>Salmonella bongori</i>
XN5VC	<i>Salmonella enterica</i>
XN4AM	<i>Salmonella</i> Typhi
XN0UV	<i>Salmonella</i> Paratyphi
XN1K5	<i>Salmonella</i> Paratyphi A
XN322	<i>Salmonella</i> Paratyphi B
XN5TR	<i>Salmonella</i> Paratyphi C
1A07	Typhoid fever
1A08	Paratyphoid fever
MG50.9	Antibiotic resistant <i>Salmonella</i>

³ Within pathogen species and groups, there are likely to be variations in resistance, which could be further impacted by characteristics of the water supply and operating conditions. Resistance is based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: 99% inactivation at 20 °C where, generally, Low represents a Ct₉₉ of < 1 min.mg/L, Moderate 1–30 min.mg/L and High > 30 min.mg/L (where C = the concentration of free chlorine in mg/L and t = contact time in minutes) under the following conditions: the infective stage is freely suspended in water treated at conventional doses and contact times, and the pH is between 7 and 8.

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Annex. Approach to content development and declarations of interest

The background documents were developed drawing from the following resources (i) microbial fact sheets from the fourth edition of the WHO Guidelines for Drinking-Water Quality (ii) pathogen-specific chapters in the Global Water Pathogen Project (GWPP) book “*Sanitation and Disease in the 21st Century*”, an update of the seminal book by Feachem et al. 1983 “*Sanitation and Disease: Health Aspects of Excreta and Wastewater Management*” and (iii) a rapid review of the literature at the start of the development of the background documents and before finalization in December 2023, focusing on identifying additional recent high quality literature reviews on a given pathogen. The documents were prepared by members of the WHO drinking-water quality microbiology group as well as external scientists with expertise in the relevant pathogens. The documents went through several iterations based on inputs from members of the WHO drinking-water quality microbiology working group and WHO technical staff. In addition, each document was reviewed by at least two external peer reviewers including authors of the GWPP book chapters. The extensive review process was designed to ensure technical accuracy and to identify any data gaps. The background documents were discussed at three expert meetings to resolve any issues and ensure the content reflected a consensus of expert opinions. Upon satisfactory resolution of any further data gaps or issues, the background documents went through technical editing before finalization.

Before the key expert meetings mentioned above, and for this work more broadly, all microbiology working group members submitted declarations of interest to WHO. These were to disclose any potential conflicts of interest that might affect, or might reasonably be perceived to affect, their objectivity and independence in relation to the subject matter of the guidance. WHO reviewed each of the declarations and concluded that none could give rise to a potential or reasonably perceived conflict of interest related to the subjects discussed at the meetings or covered by the guidance.